



OSCILLATIONS, WAVES AND OPTICS

PHYSICS FOR CHEMISTRY STUDENTS

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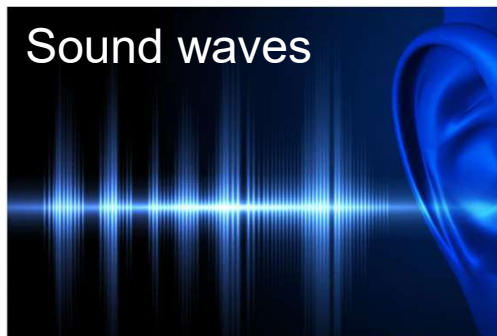
课程简介

本课程的是经典物理系列课程的一部分，面向理工科，特别是化学专业的本科二年级学生开设。课程的主要内容包括振动，波，光学原理，与传统的教学不同，本课程将把物理中有关振动、波动以及光学原理的部分进行介绍和对比。强调物理概念和思想的理解，物理规律形成的背景和应用，强化数学工具的使用原理，淡化其中繁琐的计算过程，旨在全面且高效的打下理工类学生必要的振动与光学物理基础。

参考教材：《大学物理学》，张三慧，清华大学出版社

THE PHYSICS OF WAVES, by HOWARD GEORGI
Originally published by PRENTICE HALL Englewood Cliffs

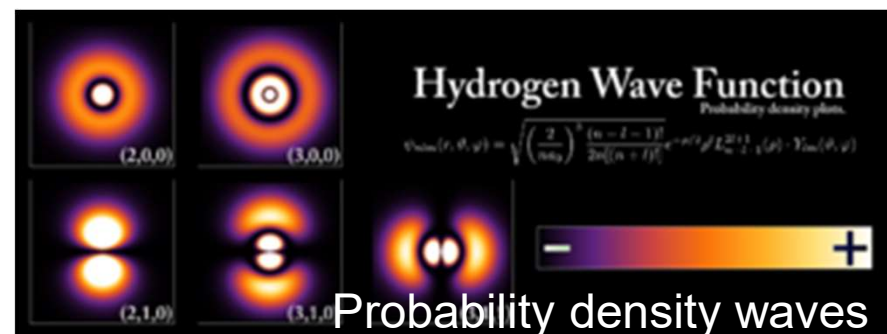
INTRODUCTION



Classic

Quantum

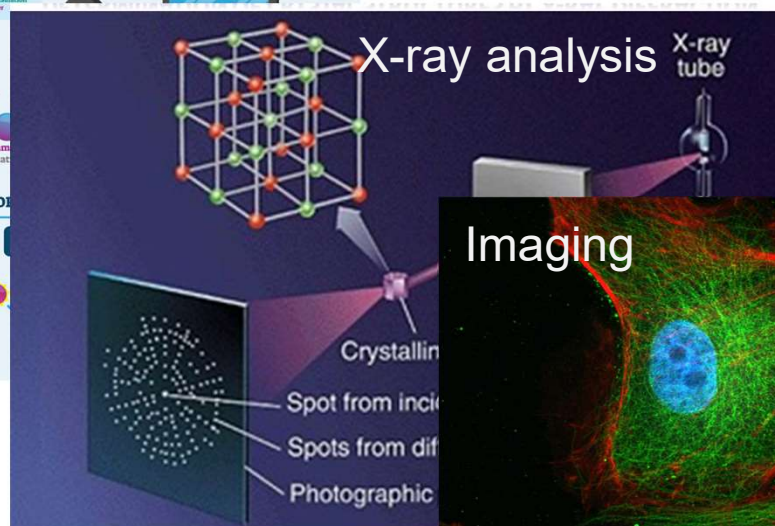
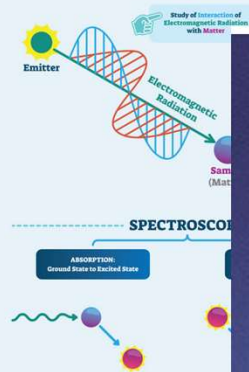
Electromagnetics
Optics
Solid physics
Quantum physics
...



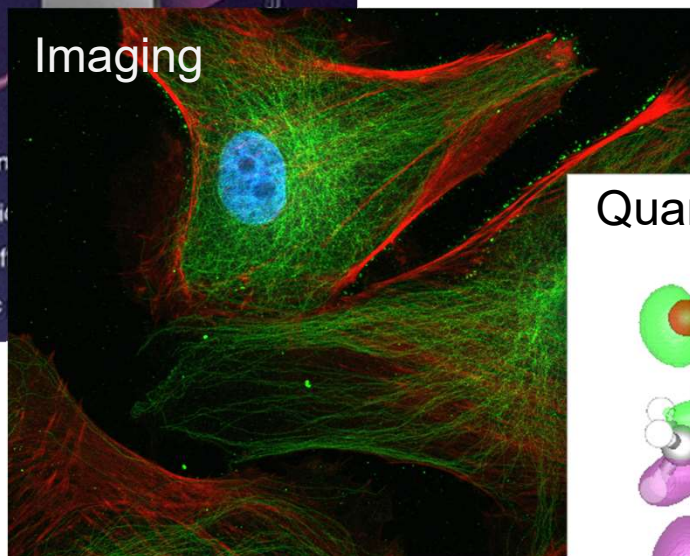
Probability density waves

FOR CHEMISTS

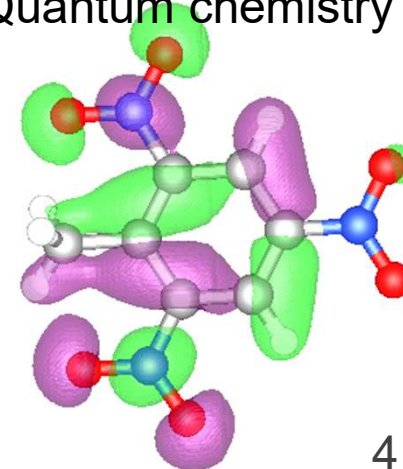
WHAT IS SPECTROSCOPY?



Imaging



Quantum chemistry



CONTENT*

Chapter 1. Oscillation

- 1.1 Harmonic oscillator (简谐振动) 【1】
- 1.2 Damped oscillator(阻尼振动) 【1】
- 1.3 Forced oscillator (受迫振动) 【2】
- 1.4 Resonance and Q factor (共振与品质因子) 【2】

Chapter 2. Coupled oscillation and Normal mode

- 2.1 Coupled oscillators (耦合振动) 【3】
- 2.2 Normal mode (简振模) 【4】
- 2.3 Molecular vibrations* 【4】

CONTENT*

Chapter 3. Waves

3.1 Wave equation (波动方程) 【5】

3.2 Standing and travelling waves (驻波与行波) 【6-7】

3.3 Longitudinal wave and sound (纵波与声波) 【8】

3.4 Fourier series* (傅里叶级数) 【9】

Chapter 4. Light and Wave optics

4.1 Superposition of Electromagnetic waves (电磁波的叠加) 【10】

4.2 Coherence & Interference (相干性与干涉) 【11】

4.3 Interferometers (干涉仪) 【12】

4.4 Far field diffraction (衍射) 【13】

4.5 Polarization (偏振) 【14】

4.6 Nonlinear optics and Laser* (非线性光学和激光原理) 【15】



DEMONSTRATION

- Harmonic oscillator & Damped oscillator
- Resonance
- Coupled oscillation & Normal mode



EXAMINATION & GRADING

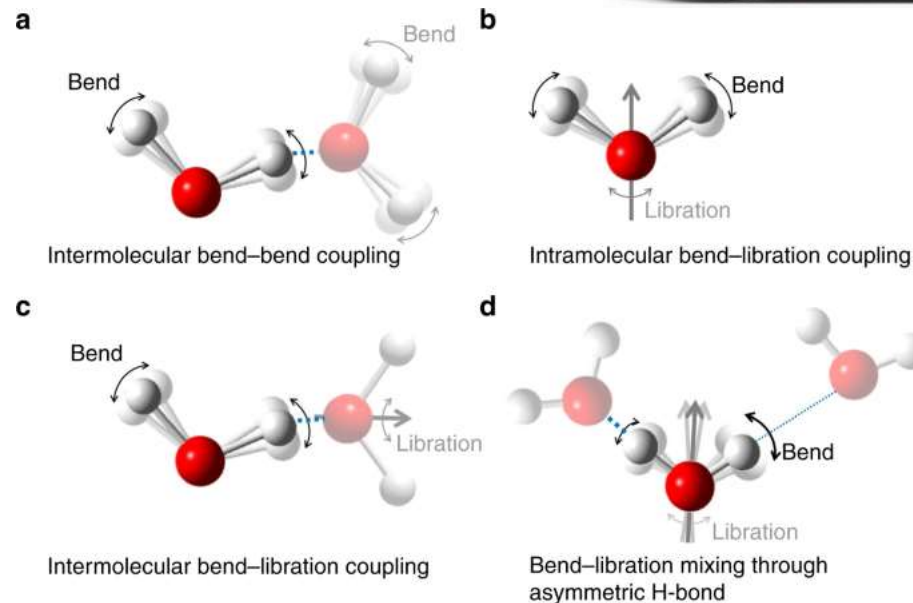
Mini projects + quiz: 70%

Final: 30%

CHAPTER 1. OSCILLATION

1.1. HARMONIC OSCILLATION

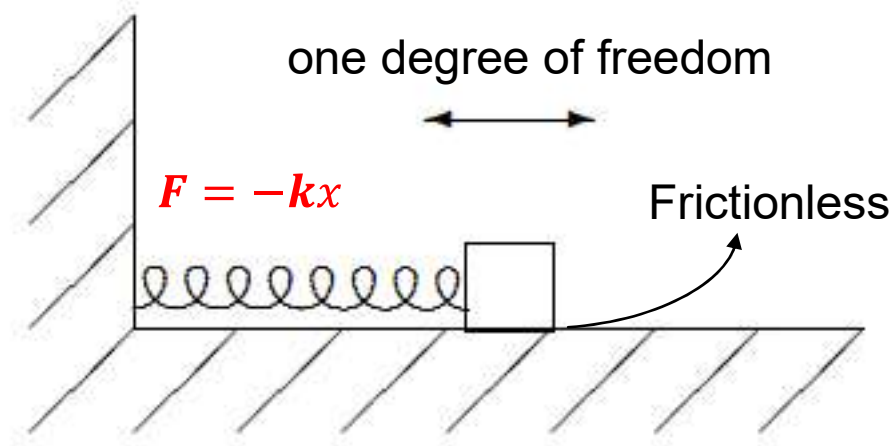
Harmonic oscillators:



CHAPTER 1. OSCILLATION

1.1. HARMONIC OSCILLATION

Hooke's Law



$$m\ddot{x} = -kx(t)$$

Figure 1.1: A mass on a spring.

Newton's Law

$$F = m\ddot{x}$$

CHAPTER 1. OSCILLATION

1.1. HARMONIC OSCILLATION

Equation of motion: $m\ddot{x}(t) = -kx(t)$ $\ddot{x}(t) = -k/m \cdot x(t)$

$$k/m = \omega^2$$

$$\sqrt{k/m} = \omega = 2\pi\nu$$

(angular frequency)

General solution: $x(t) = a\cos(\omega t) + b\sin(\omega t)$

CHAPTER 1. OSCILLATION

1.1. HARMONIC OSCILLATION

Equation of motion: $\ddot{x}(t) = -\omega^2 x(t)$

General solution: $x(t) = a \cos(\omega t) + b \sin(\omega t)$

$$A = \sqrt{a^2 + b^2}$$

$$\sin \phi = a / \sqrt{a^2 + b^2}$$

$$\cos \phi = b / \sqrt{a^2 + b^2}$$



$$A = \sqrt{a^2 + b^2}$$

$$\sin \varphi = -b / \sqrt{a^2 + b^2}$$

$$\cos \varphi = a / \sqrt{a^2 + b^2}$$

$$x(t) = A \sin(\omega t + \phi)$$

$$x(t) = A \cos(\omega t + \varphi)$$

$$\phi = \varphi + \frac{\pi}{2}$$

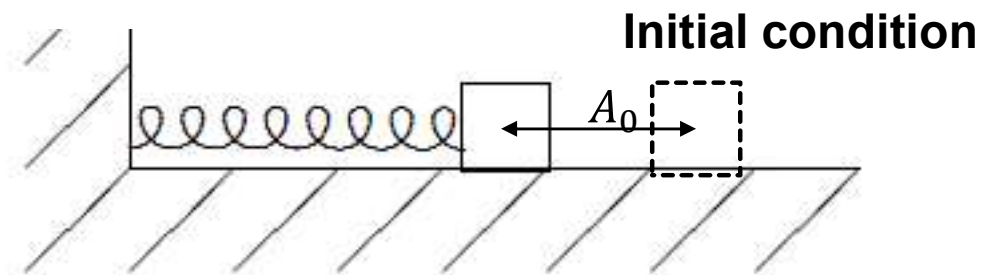
$$x(t) = \text{imag}[Ae^{-i(\omega t + \phi)}]$$

$$x(t) = \text{Real}[Ae^{-i(\omega t + \varphi)}]$$

CHAPTER 1. OSCILLATION

1.1. HARMONIC OSCILLATION

General solution: $x(t) = A \cos(\omega t + \varphi)$



$$\dot{x}(0) = 0 \quad \Rightarrow \quad -A\omega \sin \varphi = 0 \quad \varphi = 0$$

$$x(0) = A_0 \quad \Rightarrow \quad A \cos \varphi = A_0 \quad A = A_0$$

$$x(t) = A_0 \cos(\omega t)$$

CHAPTER 1. OSCILLATION

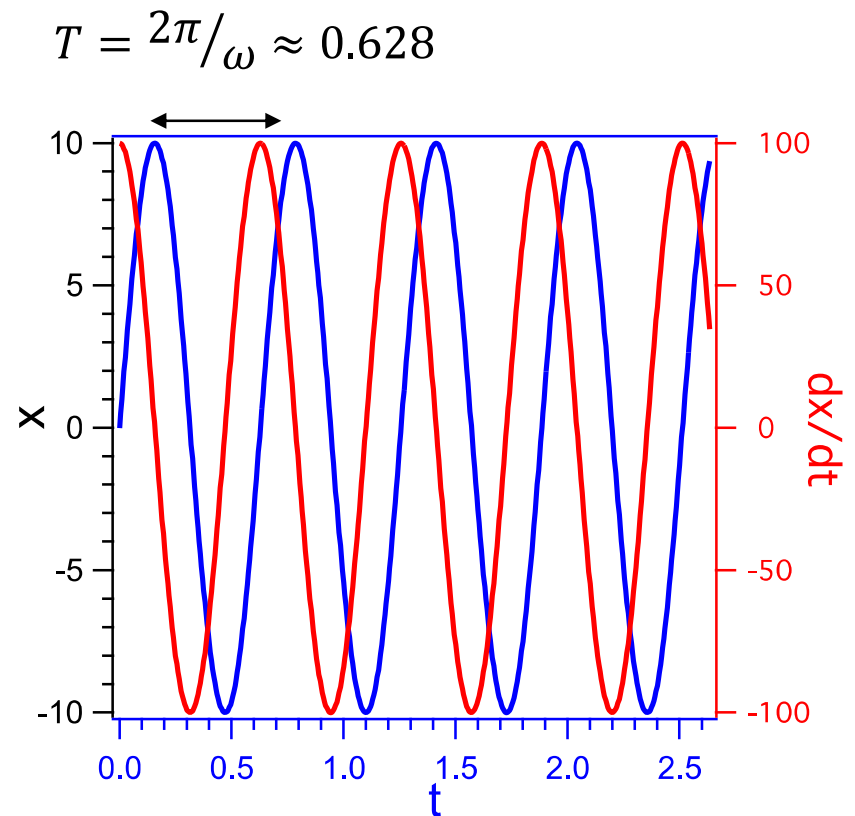
1.1. HARMONIC OSCILLATION

$$A_0 = 10$$

$$\omega = \sqrt{k/m} = 10 \quad \varphi = 0$$

$$x(t) = A_0 \cos(\omega t)$$

$$\begin{aligned} \dot{x}(t) &= -A_0 \omega \sin(\omega t) \\ &= A_0 \omega \cos(\omega t - \pi/2) \end{aligned}$$



CHAPTER 1. OSCILLATION

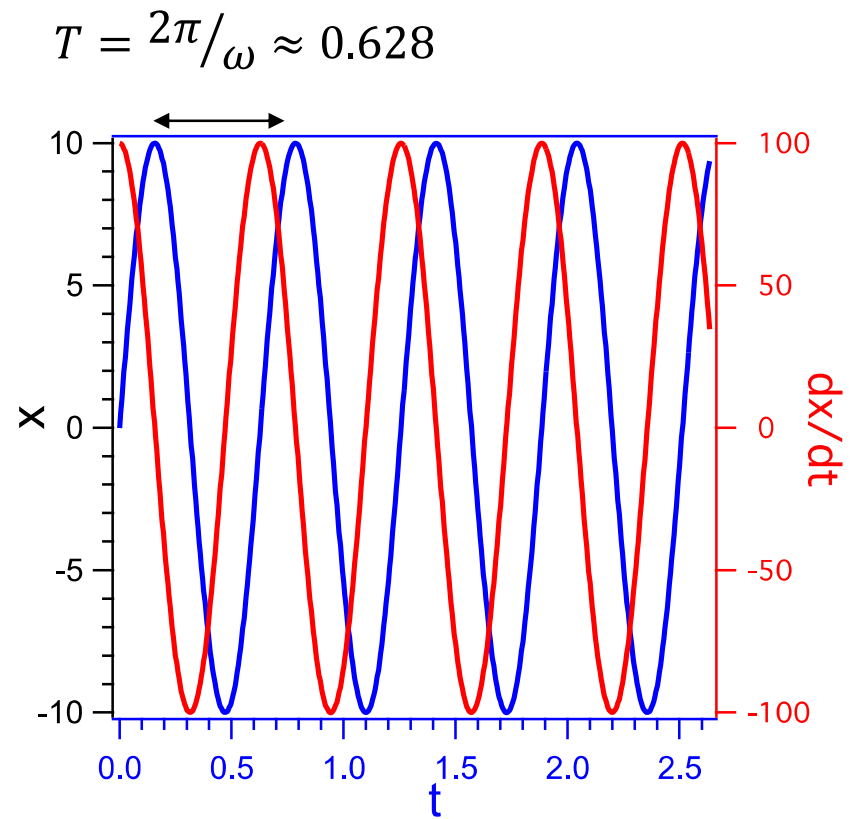
1.1. HARMONIC OSCILLATION

$$x(t) = A \cos(\omega t + \varphi)$$

Amplitude

Angular frequency
intrinsic ($\sqrt{k/m}$)

Phase



CHAPTER 1. OSCILLATION

1.1. HARMONIC OSCILLATION

$$x(t) = A \cos(\omega t + \varphi)$$

Kinetic Energy: $E_k = \frac{1}{2} m \dot{x}^2 = \frac{1}{2} m \omega^2 A^2 \sin^2(\omega t + \varphi)$

Potential Energy: $V = \frac{1}{2} k x^2 = \frac{1}{2} k A^2 \cos^2(\omega t + \varphi)$

$$\omega = \sqrt{k/m}$$

Total Energy: $V + E_k = \frac{1}{2} k A^2$

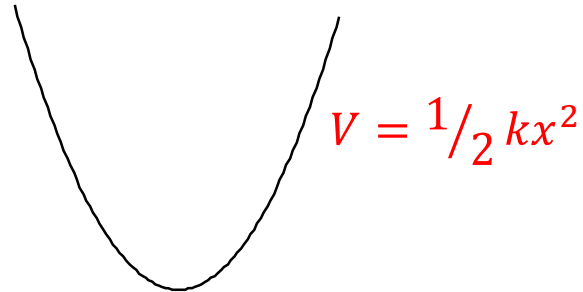
Energy conservation

CHAPTER 1. OSCILLATION

1.1. HARMONIC OSCILLATOR

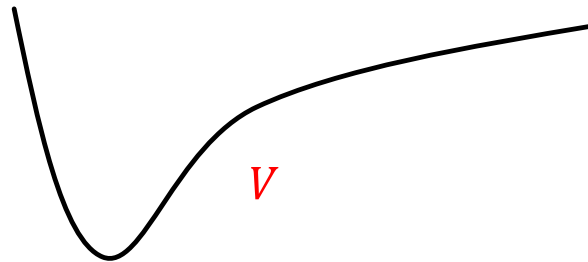
Hooke's Law (ideal)

$$F = -kx$$



Non-idea case:

$$F(x) = -\frac{\partial V}{\partial x}$$



CHAPTER 1. OSCILLATION

1.1. HARMONIC OSCILLATION

Energy:

$$F(x) = m\ddot{x} = -\frac{\partial V}{\partial x}$$

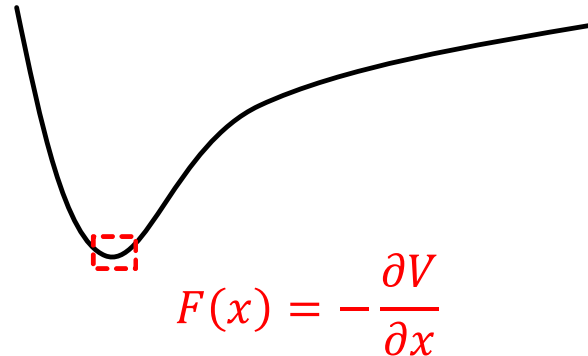
$$\begin{aligned} & \frac{d}{dt} \left(\frac{1}{2} m \dot{x}^2(t) + V(x) \right) \\ &= m \dot{x}(t) \ddot{x}(t) + \frac{\partial V(x)}{\partial x} \frac{dx(t)}{dt} \\ &= \dot{x}(t) \left[m \ddot{x}(t) + \frac{\partial V(x)}{\partial x} \right] = 0 \end{aligned}$$

Energy conservation

CHAPTER 1. OSCILLATION

1.1. HARMONIC OSCILLATOR

Non-idea case:



Taylor expansion adjacent to $x=0$:

$$V(x) = V(0) + V'(0)x + \frac{1}{2!}V''(0)x^2 + \dots$$

$$F(x) = -V'(0) - V''(0)x + \dots$$

$$F(x) \approx -V''(0)x$$

Hooke's Law

Many small oscillations could be treated “harmonically”

1.1. HARMONIC OSCILLATION

How did we solve the problem?

1. Force analysis (Physics)
2. Equation of motion (Math)
3. General solution (Math)
4. Boundary condition(Physics)

CHAPTER 1. OSCILLATION

1.1. HARMONIC OSCILLATION

LC circuit

Force analysis

$$V_C + V_L = 0$$

$$I_C = I_L$$

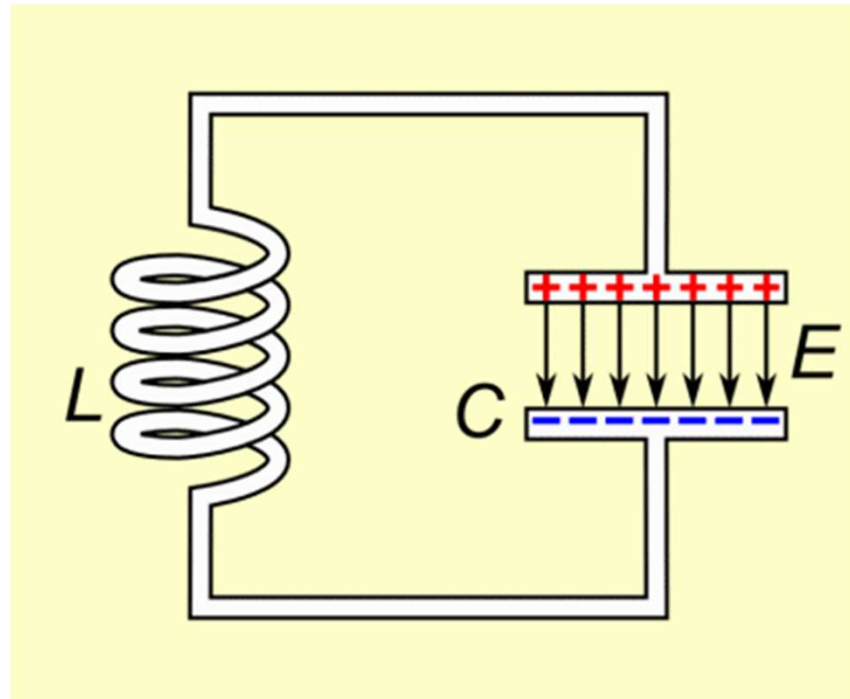
$$L \cdot dI_L/dt = -V_L$$

$$C \cdot dV_C/dt = I_C$$

$$LC \cdot d^2 I_L/dt^2 = I_C$$

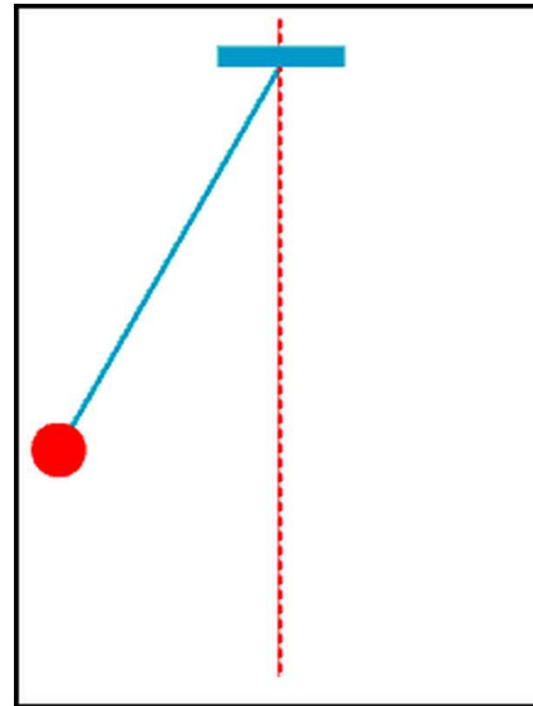
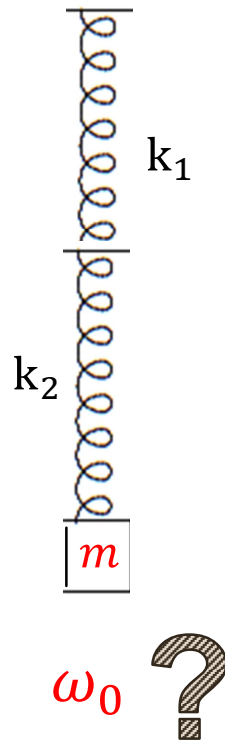
$$\ddot{I}_L + \frac{1}{LC} I_L = 0$$

Harmonic Oscillator



CHAPTER 1. OSCILLATION

HOMework

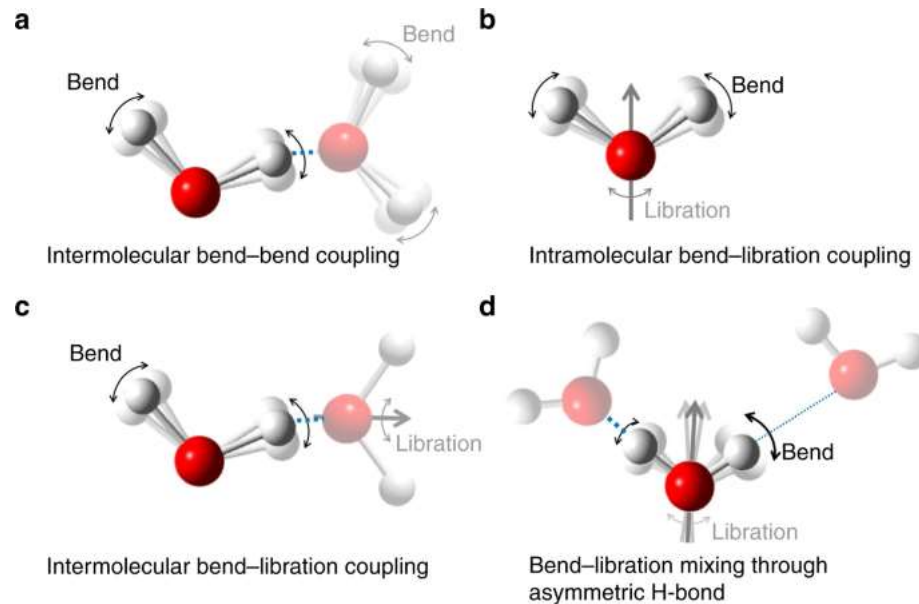


CHAPTER 1. OSCILLATION

1.2. DAMPED OSCILLATOR



Oscillate forever ???



CHAPTER 1. OSCILLATION

1.2. DAMPED OSCILLATOR

Hooke's Law

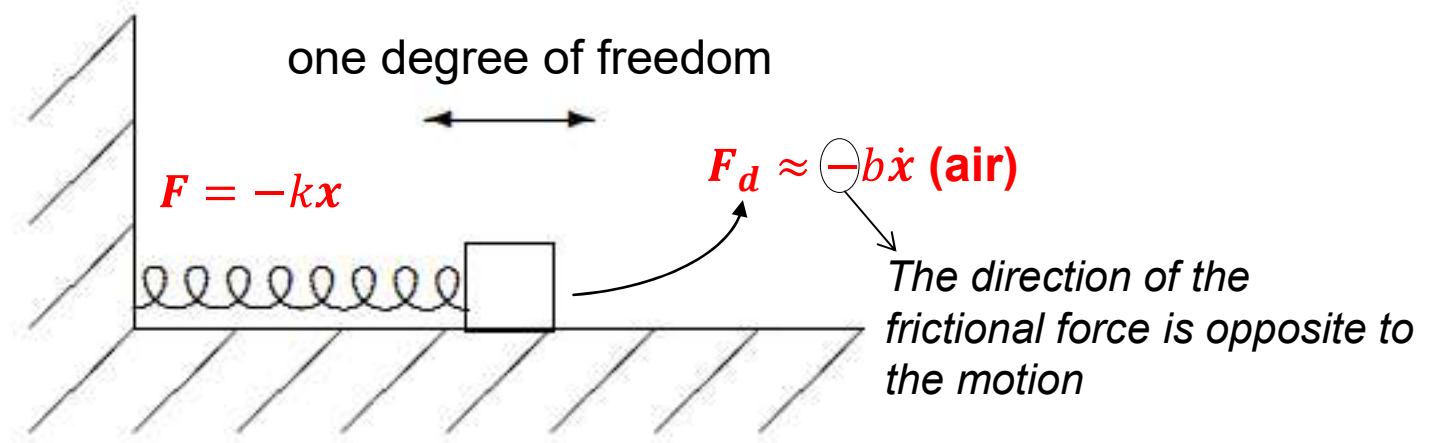


Figure 1.1: A mass on a spring.

Equation of motion

$$m\ddot{x} = -kx - b\dot{x}$$

This force is always pointing to the equilibrium position

CHAPTER 1. OSCILLATION

1.2. DAMPED OSCILLATOR

Equation of motion

$$m\ddot{x} = -kx - b\dot{x}$$

$$\gamma = b/m$$

$$k/m = \omega_0^2$$



$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x(t) = 0$$

linear 2nd order homogeneous ODE

Let

$$x(t) = e^{\alpha t}$$

$$\alpha^2 e^{\alpha t} + \gamma \alpha e^{\alpha t} + \omega_0^2 e^{\alpha t} = 0$$



$$\alpha^2 + \gamma\alpha + \omega_0^2 = 0$$

Characteristic equation

CHAPTER 1. OSCILLATION

1.2. DAMPED OSCILLATOR

Equation of motion

$$m\ddot{x} = -kx - b\dot{x}$$

Characteristic equation

$$\alpha^2 + \gamma\alpha + \omega_0^2 = 0$$

CHAPTER 1. OSCILLATION

1.2. DAMPED OSCILLATOR

Trial solution:

$$x(t) = e^{\alpha t}$$

$$\alpha = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega_0^2}}{2}$$

Case 1: Drag force is large

过阻尼

$$\gamma^2 > 4\omega_0^2$$

$$\gamma^2 - 4\omega_0^2 > 0$$

Let

$$\gamma^2 - 4\omega_0^2 = 4\omega^2$$

CHAPTER 1. OSCILLATION

1.2. DAMPED OSCILLATOR

Case 1:

General solution:

$$x(t) = ae^{-\frac{\gamma}{2}t} e^{\omega t} + be^{-\frac{\gamma}{2}t} e^{-\omega t}$$

$$\alpha = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega_0^2}}{2} = \frac{-\gamma \pm 2\omega}{2}$$

Drag force is large

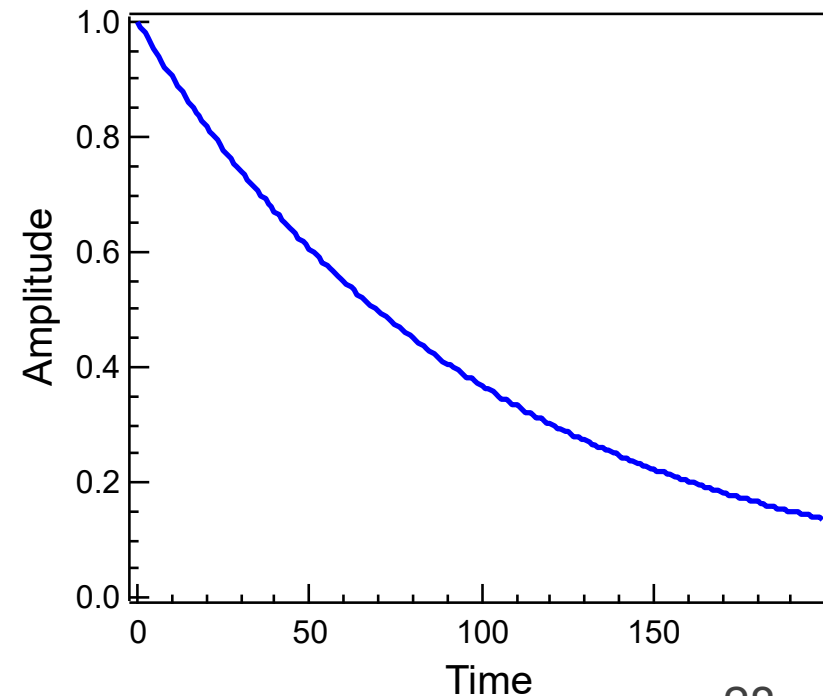
$$\gamma^2 > 4\omega_0^2$$

$$f_d \approx -b\dot{x}$$

$$\gamma = b/m$$

$$k/m = \omega_0^2$$

$$\gamma^2 - 4\omega_0^2 > 0$$



CHAPTER 1. OSCILLATION

1.2. DAMPED OSCILLATOR

$$\ddot{x} + \gamma \dot{x} + \omega_0^2 x(t) = 0$$

$$x(t) = e^{\alpha t}$$

$$\alpha = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega_0^2}}{2}$$

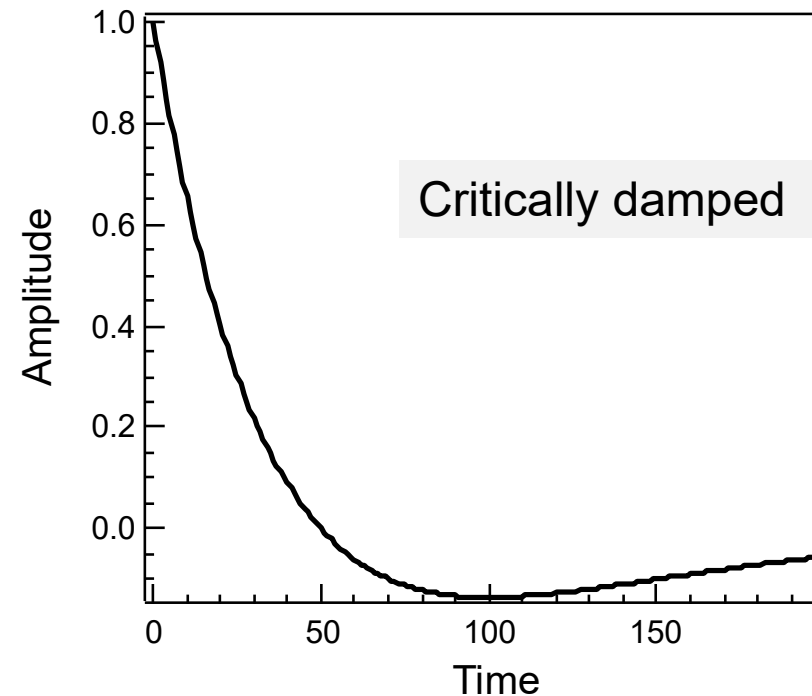
Case 2: $\gamma^2 - 4\omega_0^2 = 0$

$$x_1(t) = ae^{-\frac{\gamma}{2}t}$$

$$x_2(t) = bte^{-\frac{\gamma}{2}t}$$

General solution:

$$x(t) = (a + bt)e^{-\frac{\gamma}{2}t}$$



CHAPTER 1. OSCILLATION

1.2. DAMPED OSCILLATOR

Case 3 (drag force is small): $\gamma^2 - 4\omega_0^2 < 0$ 欠阻尼

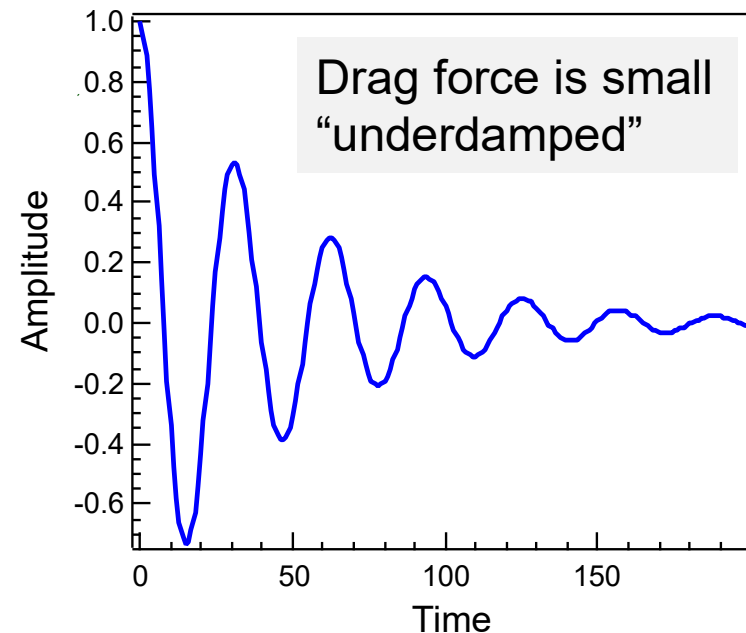
$$\sqrt{\gamma^2 - 4\omega_0^2} = i2\omega \quad \alpha = \frac{-\gamma \pm i2\omega}{2}$$

$$x_1(t) = e^{\frac{-\gamma + i2\omega}{2}t}$$

$$x_2(t) = e^{\frac{-\gamma - i2\omega}{2}t}$$

General solution:

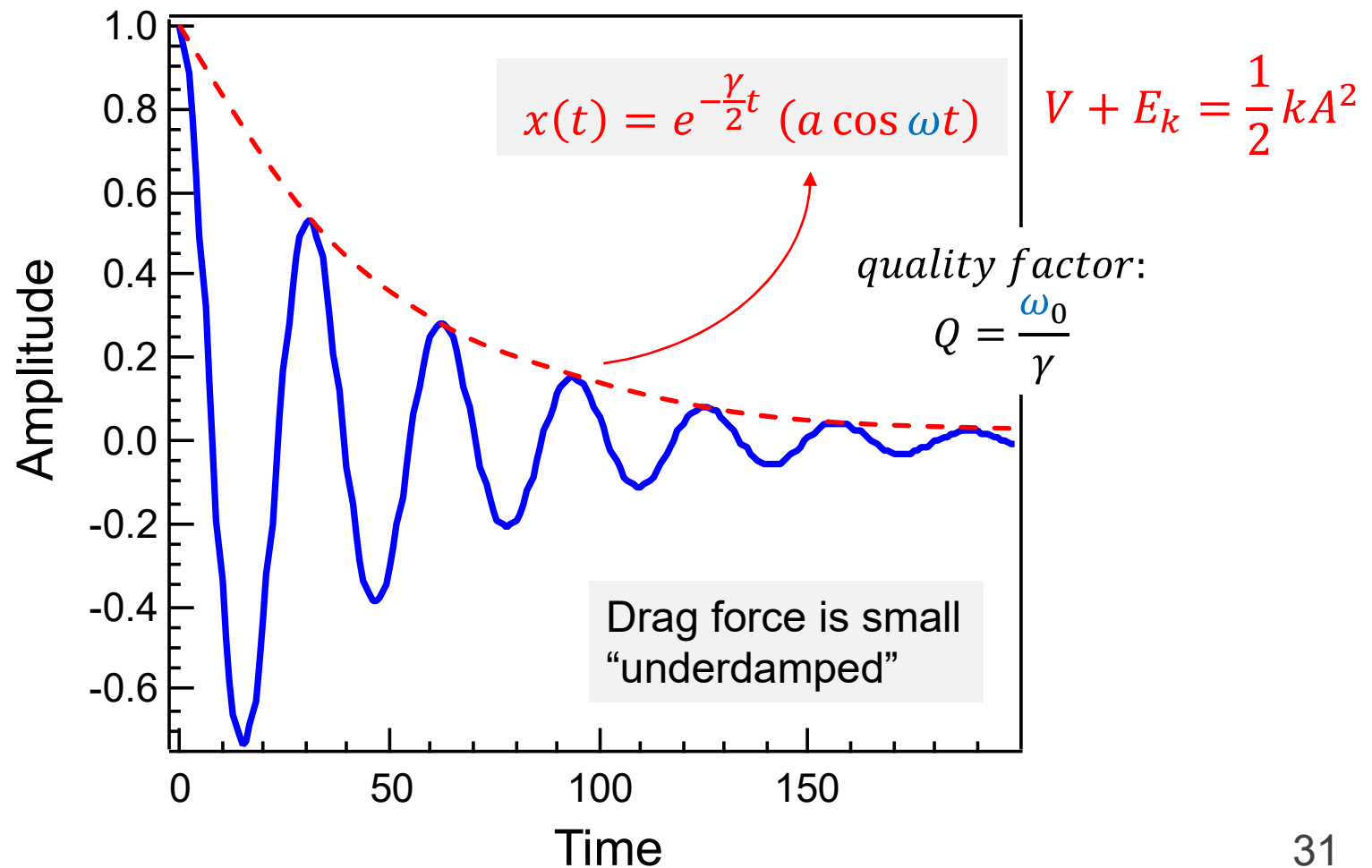
$$x(t) = e^{-\frac{\gamma}{2}t} (a \cos \omega t + b \sin \omega t)$$
$$x(t) = Ae^{-\frac{\gamma}{2}t} \sin(\omega t + \phi)$$



CHAPTER 1. OSCILLATION

1.2. DAMPED OSCILLATOR

Case 3:



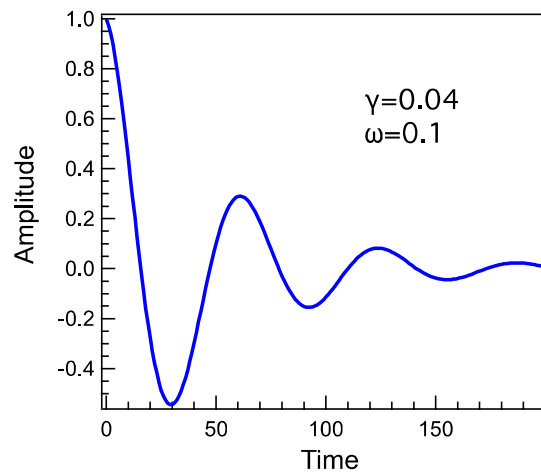
CHAPTER 1. OSCILLATION

1.2. DAMPED OSCILLATOR

$$x(t) = e^{-\frac{\gamma}{2}t} (a \cos \omega t)$$

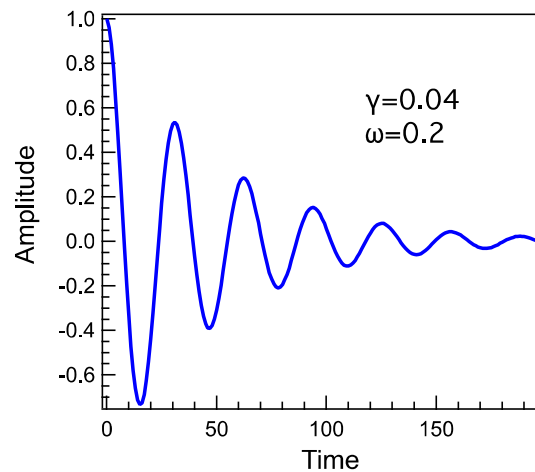
quality factor:

$$Q = \frac{\omega_0}{\gamma} = 2.5$$



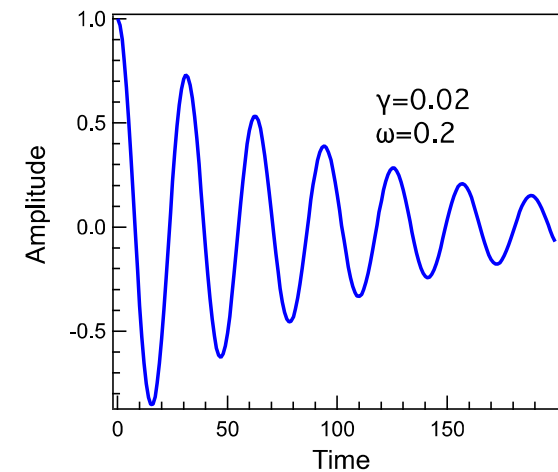
quality factor:

$$Q = \frac{\omega_0}{\gamma} = 5$$



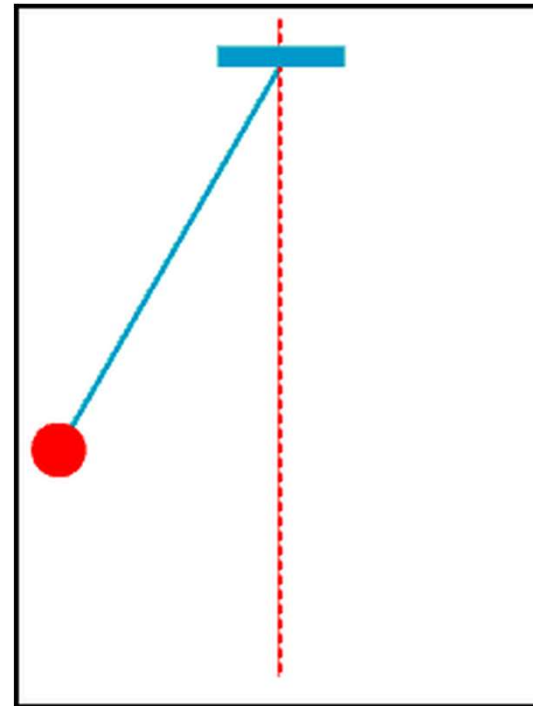
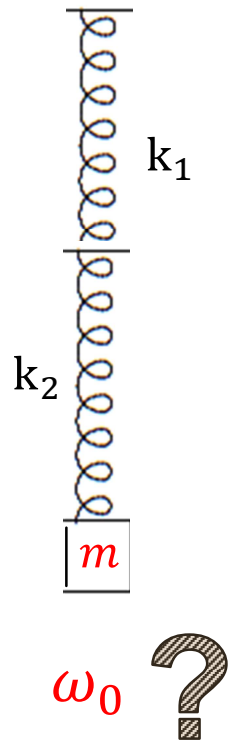
quality factor:

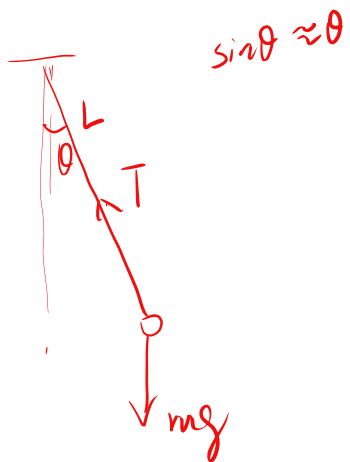
$$Q = \frac{\omega_0}{\gamma} = 10$$



CHAPTER 1. OSCILLATION

HOMework





$$x: m\ddot{x} = -mg \tan \theta \approx -mg \theta = -mg \frac{x}{L}$$

$$\ddot{x} = -g \frac{x}{L}$$

$$\ddot{x} + \frac{g}{L} x = 0$$

$$x^2 + \frac{g}{L} = 0 \quad x = \pm i \sqrt{\frac{g}{L}}$$

$$C_1 e^{i\sqrt{\frac{g}{L}}x} + C_2 e^{-i\sqrt{\frac{g}{L}}x}$$

$$A \sin\left(\sqrt{\frac{g}{L}}x + p\right)$$